

WELL COMPLETION REPORT (SYNTHETIC)

OIL-NHK-003 - Nahorkatiya-003

NWIS prototype document - not an Oil India Limited operational record

API Number	OIL-ASM-000026
Field / District	Nahorkatiya / Dibrugarh
Block	PEL-ASSAM-02
Basin	Assam-Arakan
Formation	Barail (BAR)
Spud / Completion	2023-06-09 / 2023-09-21
Measured / TVD	2691 m / 2663 m
Trajectory	Vertical; inc 0.8 deg; az 176.4 deg
Formation Interval	1629-2320 m
Lithology	Interbedded sandstone and shale
Pore Pressure Reference	3028 psi
Fracture Gradient	15.56 ppg
Mud System	KCl Polymer WBM
NWIS Well Risk	High (59.5/100)

Casing Program

String	Set Depth (m)	Purpose
20 in conductor	80	Conductor
13 3/8 in casing	672	Surface
9 5/8 in casing	1560	Intermediate
7 in liner/casing	2368	Production

Significant Historical Events and Lessons

Date / Depth	Event	Severity	Evidence / mitigation / lesson
2023-06-20 405 m	Stuck Pipe	High	High drag was recorded while circulating remained possible. Mitigation: Condition mud and verify cuttings transport before connection. Lesson: Rising torque plus falling ROP should trigger hole-cleaning action.
2023-06-28 635 m	Kick	High	Flow-out exceeded flow-in and gas units increased. Mitigation: Increase mud weight within fracture-margin limits and monitor gas trend. Lesson: Maintain mud-weight window using offset pressure evidence.
2023-07-08 900 m	Mud Loss	High	Partial returns observed at shale shaker. Mitigation: Stage circulation and monitor pit volume before resuming drilling. Lesson: Track flow balance and pit volume trend before losses become severe.
2023-07-11 958 m	Torque Spike	High	Torque rose without corresponding pressure anomaly. Mitigation: Ream and condition hole if torque remains elevated. Lesson: Use offset torque trend to set adaptive operating envelope.
2023-07-11 973 m	Pressure Spike	High	Standpipe pressure increased rapidly at nearly stable flow. Mitigation: Reduce pump rate, verify restriction and restore circulation gradually. Lesson: Use depth-matched offset hydraulic baseline for early warning.
2023-07-17 1116 m	Mud Loss	High	Flow-out dropped below flow-in with pit loss. Mitigation: Stage circulation and monitor pit volume before resuming drilling. Lesson: Track flow balance and pit volume trend before losses become severe.
2023-07-31 1467 m	Cementing Issue	High	Observed returns did not match calculated displacement. Mitigation: Control displacement rate and reconcile pumped/returned volumes. Lesson: Offset loss history should inform cementing hydraulics.

Completion Summary

The synthetic well reached 2691 m MD in the Barail interval. The NWIS knowledge record links 19 historical operational events to depth-matched telemetry. Risk classification is derived from formation, depth, trajectory and event history rather than formation alone.

This report is generated solely for SIH/NWIS prototyping and model development.